

UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS
Doctoral Program in Deep Tech with a Focus on Artificial Intelligence
and Emerging Technologies

DELIVERABLE No. 2 – METHOD FIT MATRIX (E.D.F.C.V.)

Research topic: Artificial intelligence governance maturity model for Peruvian organizations within the framework of Law No. 31814 and NTP-ISO/IEC 42001:2025: design, validation, and automated compliance assessment.

Refined research question: How can a context-specific AI governance maturity model and an LLM-assisted compliance assessment artifact be designed and rigorously evaluated to help Peruvian public and private organizations assess and improve their governance maturity and effective compliance capacity?

Primary paradigm: Design Science Research (DSR), with mixed-method evidence used in support of artifact design and evaluation.

Weights: E=30% D=20% F=15% C=20% V=15%

Gate rule: Any candidate scoring $E \leq 2$ is eliminated regardless of its weighted total.

Candidate method	E 30%	D 20%	F 15%	C 20%	V 15%	Weighted score /5	Method fit defense
Design Science Research	5	4	4	5	5	4.65	Natural fit with the paradigm and the intended contribution: a maturity model, validated instrument, and LLM-based artifact. Required data are obtainable through literature, experts, organizations, and governance documents. Feasibility is demanding but manageable within the doctoral horizon. Strong fit with information-systems and AI-governance publications venues.
Survey Research	3	4	5	3	4	3.65	Viable for establishing a multi-organization maturity baseline and testing the assessment instrument. It is feasible and scalable, but it primarily produces population-level evidence; by itself it does not construct and iterability evaluate the governance model or automated assessment artifact.
Case Study	3	3	3	2	4	2.95	Viable for deep contextual understanding of AI-governance practices and documentary evidence in selected organizations. However, access to rich internal documentation may be uneven, and contextual depth alone does not deliver the principal artifact-building contribution required by the research question.

Defended choice. Design Science Research is selected because the doctoral contribution is fundamentally artifact-centerer: the study must build and evaluate a Peruvian AI-governance maturity model, a validated assessment instrument, and an LLM-assisted compliance assessment artifact. Survey research remains essential as supporting evidence for validation and the national baseline, while case-study techniques may provide contextual evidence where access permits; neither replaces DSR as the method that produces the core contribution.

Weight defense. Epistemological fit receives the highest weight (30%) because method-paradigm coherences is a non-negotiable constraint for the study. Feasibility and venue fit receive the lowest weights (15% each) because they matter operationally, but they should not override epistemological coherence or the type of contribution the doctorate is designed to produce.

Open tension. The main unresolved tension is access to senior organizational informants and sufficiently complete AI-governance documentation. Uneven access could affect both the organizational baseline and the evaluation of the LLM artifact, so access and documentary completeness must be managed explicitly during the protocol design.